



*Large single sail
hung from a single spar,
called a yard, or yardarm*

*Hull built of
overlapping wooden
planks fixed to a frame*

Viking longship
Norway c.800 CE

*Oars for use
when there
is no wind*

*Oars manned by as many
as 100 crew members*

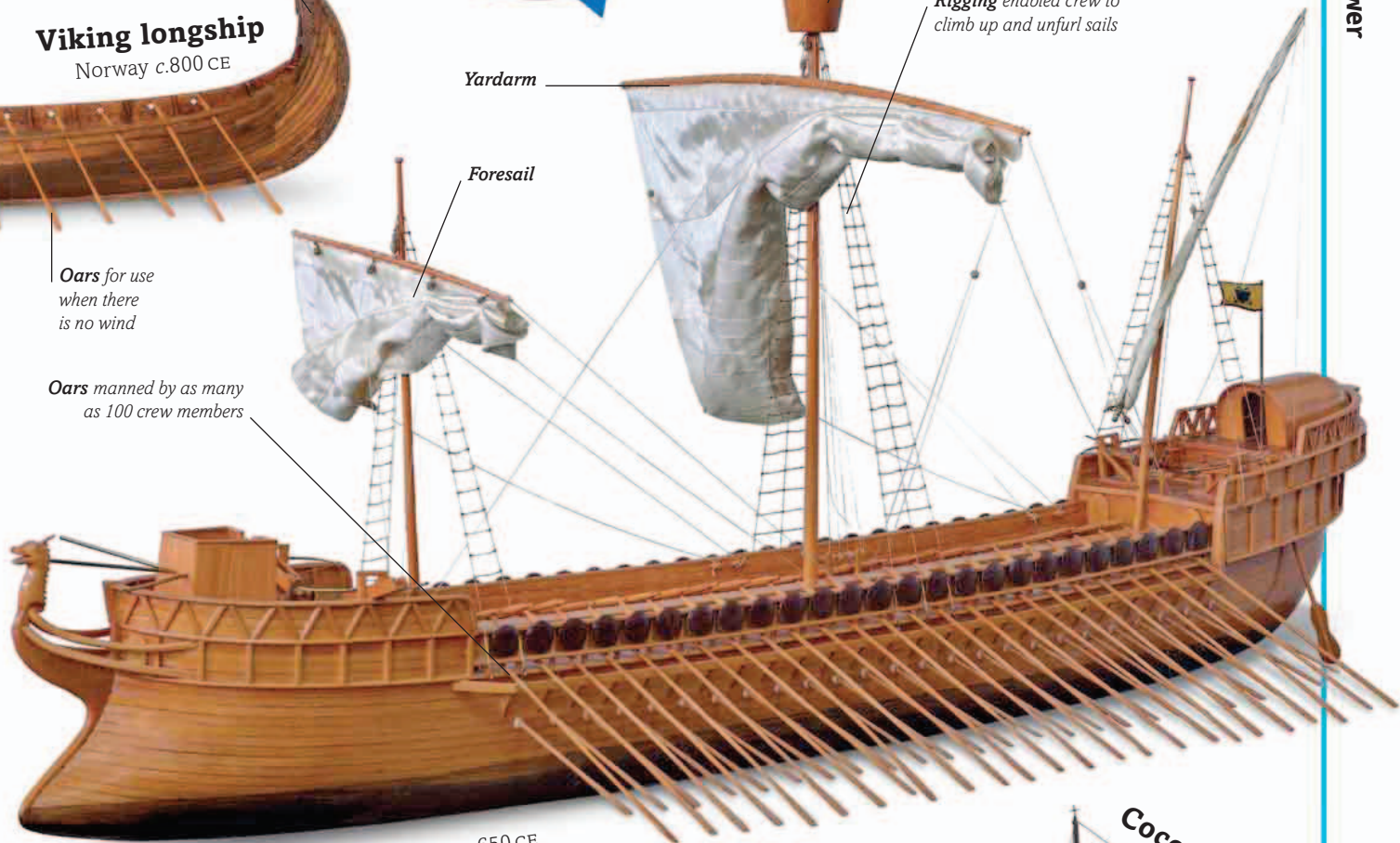
The
dragon head
on the prow of
this dromon fired
**burning
flames** at
the enemy.

*Lookout position at the top
of the main mast, for spotting
approaching ships or land*

*Rigging enabled crew to
climb up and unfurl sails*

Yardarm

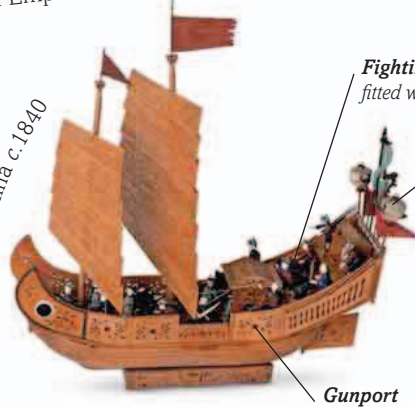
Foresail



Dromon Eastern Roman Empire c.650 CE

During
the Ming
Dynasty, China
had a navy with
more than **3,000**
war junks.

Junk China c.1840



*Fighting junk
fitted with guns*

Lantern

Wooden
rudder

Gunport

Cocca Italy c.1500



South America, and also in China, where they were often equipped to the **junks** that sailed the Pacific and Indian oceans. Many ancient sailing ships, such as **Phoenician war galleys**, **Roman galleys**, and **Viking longships**, were fitted with rows of oars, for when there was no wind.

Viking longships were designed with shallow hulls so they could sail right up to the shore to attack and raid settlements. The Vikings were skilled sailors who traveled all across Europe and, around 1000 CE, crossed the Atlantic reaching Newfoundland in Canada.